

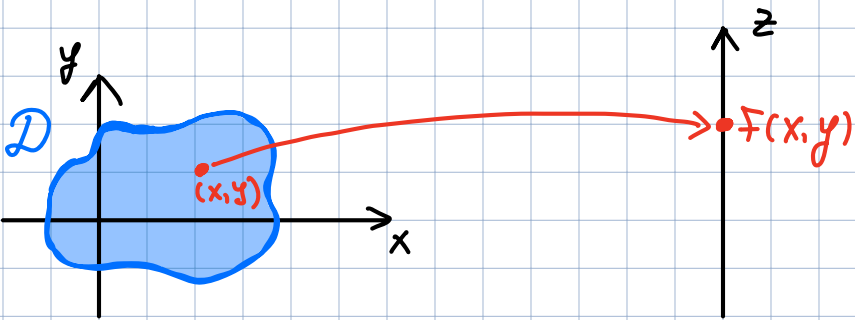
Last time: Functions of several variables

Function of two variables:

$$f: (x, y) \mapsto f(x, y)$$

in $D \subset \mathbb{R}^2$ in $\mathbb{R}$

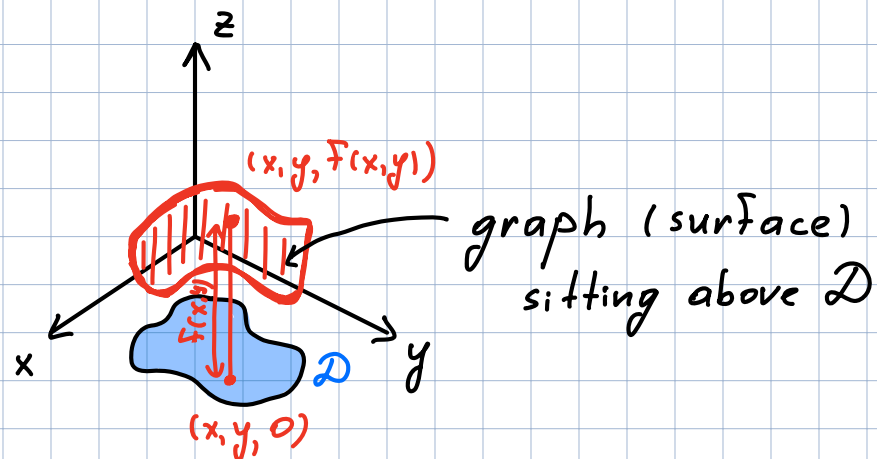
$\uparrow$ domain



range: set of all values of $f(x, y) \in \mathbb{R}$

domain: all (x, y) s.t. $f(x, y)$ is well-defined.

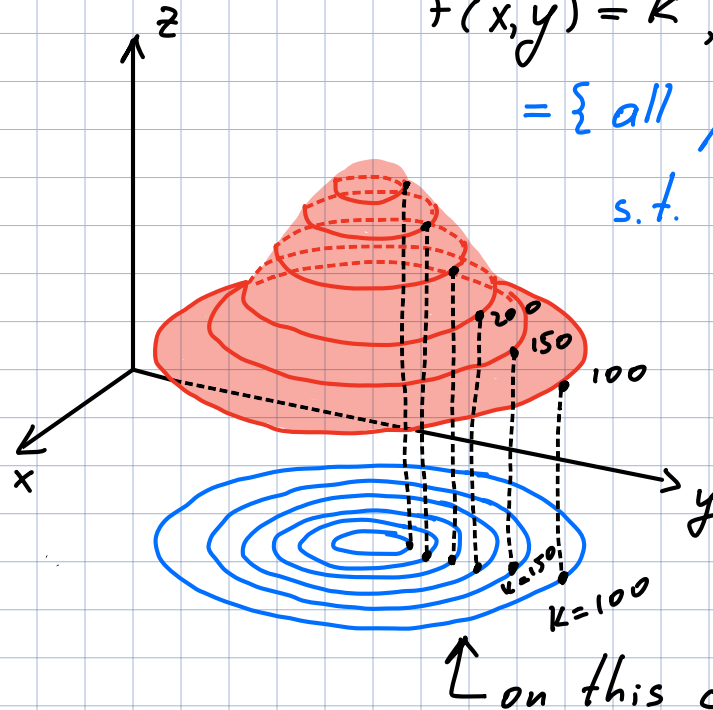
Graph: Set of points (x, y, z) in $\mathbb{R}^3$
with $(x, y) \in D$, $z = f(x, y)$



Level curves: curves with eq.

$$f(x, y) = k, \text{ i.e.}$$

$= \{ \text{all points } (x, y) \text{ s.t. } f(x, y) = k \}$

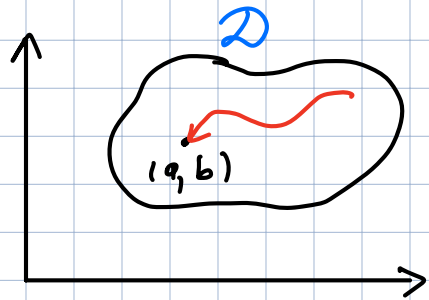


Functions of 3 variables:

$$f(x, y, z)$$

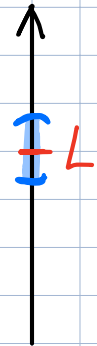
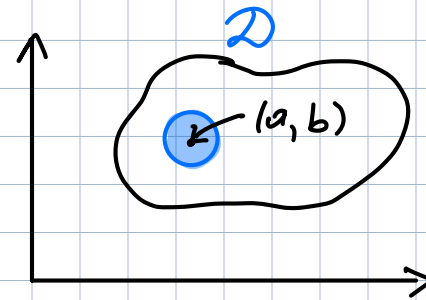
Limits and continuity:

$\lim_{(x,y) \rightarrow (a,b)} f(x,y) = L$ - The limit of $f(x,y)$ as (x,y) approaches (a,b)



• if values of $f(x,y)$ approach L as (x,y) approaches (a,b) along any path in the domain D

• we can make $f(x,y)$ as close to L as we want by taking (x,y) sufficiently close to (a,b)



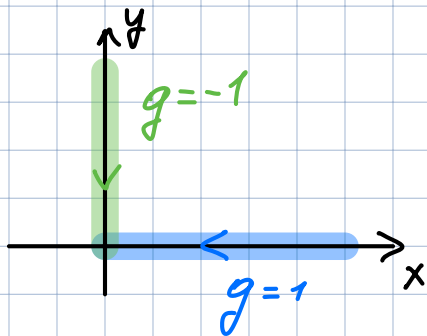
Ex: $f(x,y) = \frac{\sin(x^2+y^2)}{x^2+y^2}$

$$\lim_{(x,y) \rightarrow (0,0)} f(x,y) = 1$$

Ex: $g(x,y) = \frac{x^2 - y^2}{x^2 + y^2}$

$$\lim_{(x,y) \rightarrow 0} g(x,y) = ?$$

Sol:



approaching along x-axis g stays 1

approaching along y-axis g stays -1

$\Rightarrow$ limit as $(x,y) \rightarrow (0,0)$ does NOT exist!

• $f(x,y)$ is continuous at (a,b) if $\lim_{(x,y) \rightarrow (a,b)} f(x,y) = f(a,b)$

• f is continuous on $\mathcal{D}$ if f is continuous at each point (a,b) in $\mathcal{D}$

Ex: a polynomial: $f(x,y) = x^2 + xy + y^7$ is cont. on $\mathbb{R}^2$

a rational function: $g(x,y) = \frac{x + 2y^2}{x^2 + y^2}$ is cont. in its domain $\mathbb{R}^2 \setminus \{0,0\}$

Partial derivatives

$f(x,y)$

partial derivative w.r.t. x at (a,b) :

$$\text{set } g(x) = f(x, b)$$

← constant
↖ function
of single variable
variable

$$f_x(a,b) = g'(a)$$

likewise, partial derivative w.r.t. y at (a,b) :

$$\text{set } h(y) = f(a, y)$$

$$f_y(a,b) = h'(b)$$

I.e. to find $f_x(x,y)$, view y as a constant, differentiate w.r.t. x

to find $f_y(x,y)$, view x as a constant, differentiate w.r.t. y

Notation: $\frac{\partial f}{\partial x} = f_x(x,y)$, $\frac{\partial f}{\partial y} = f_y(x,y)$

Ex: $F(x, y) = x^3 + x^2y^3 - zy^2$

Find $F_x(2, 1)$, $F_y(2, 1)$.

Sol: $F_x(x, y) = 3x^2 + 2xy^3$

$$F_x(2, 1) = 3 \cdot 2^2 + 2 \cdot 2 \cdot 1^3 = 16$$

$$F_y(x, y) = 3x^2y^2 - 4y$$

$$F_y(2, 1) = 3 \cdot 2^2 \cdot 1^2 - 4 \cdot 1 = 8$$

Ex: $F(x, y) = \sin\left(\frac{x}{1+y}\right)$

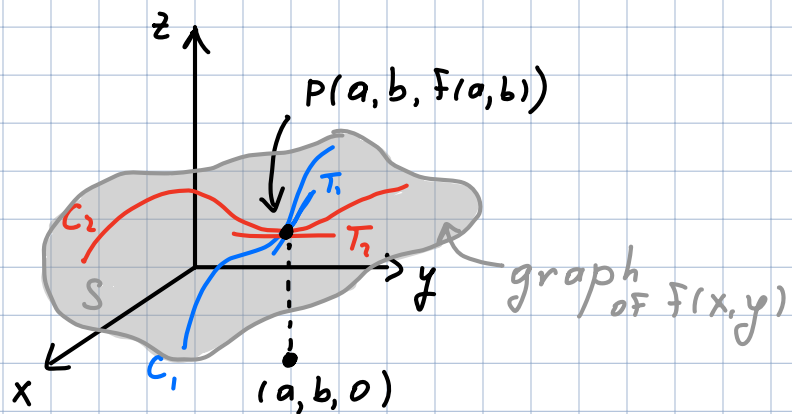
Find F_x and F_y

$$\frac{\partial F}{\partial x} = \cos\left(\frac{x}{1+y}\right) \cdot \frac{\partial}{\partial x} \left(\frac{x}{1+y}\right) = \cos\left(\frac{x}{1+y}\right) \cdot \frac{1}{1+y}$$

chain rule

$$\frac{\partial F}{\partial y} = \cos\left(\frac{x}{1+y}\right) \cdot \frac{\partial}{\partial y} \left(\frac{x}{1+y}\right) = -\cos\left(\frac{x}{1+y}\right) \frac{x}{(1+y)^2}$$

Interpretations of partial derivatives:



C_1 : graph of $g(x) = F(x, b)$
= intersection of S with plane $y = b$

C_2 : graph of $h(y) = F(a, y)$
= intersection of S with plane $x = a$

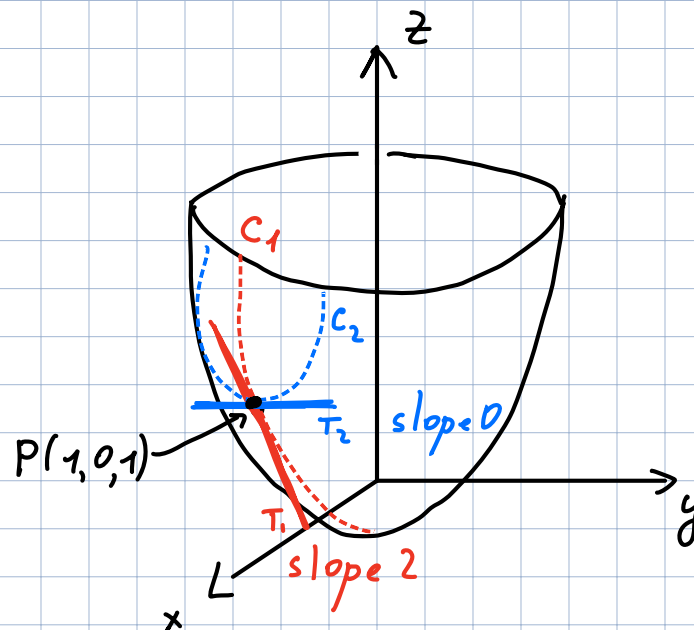
$F_x(a, b)$ = slope of tangent line T_1 to C_1 at $P(a, b, F(a, b))$

$F_y(a, b)$ = slope of tangent line T_2 to C_2 at $P(a, b, F(a, b))$

Ex: $f(x, y) = x^2 + y^2$

$$F_x(1, 0) = 2x \Big|_{(1, 0)} = 2$$

$$F_y(1, 0) = 2y \Big|_{(1, 0)} = 0$$



Higher derivatives:

$$\left. \begin{aligned} f_{xx} &= (f_x)_x = \frac{\partial^2 f}{\partial x^2}, & f_{xy} &= (f_x)_y = \frac{\partial^2 f}{\partial x \partial y}, \\ f_{yx} &= (f_y)_x = \frac{\partial^2 f}{\partial y \partial x}, & f_{yy} &= (f_y)_y = \frac{\partial^2 f}{\partial y^2} \end{aligned} \right\} \begin{array}{l} 2^{\text{nd}} \text{ order} \\ \text{derivatives} \end{array}$$

Ex: $f(x, y) = x^3 + x^2 y^3 - 2y^2$

$$f_x = 3x^2 + 2xy^3$$

$$f_y = 3x^2 y^2 - 4y$$

$$\leadsto f_{xx} = 6x + 2y^3$$

$$f_{xy} = \underline{6xy^2}$$

$$f_{yx} = \underline{6xy^2}$$

$$f_{yy} = 6x^2 y - 4$$

Rmk: $f_{xy} = f_{yx}$ - this is true for any f !

- doesn't matter in which order we are taking partial derivatives

Clairaut's Thm: $f_{xy}(a, b) = f_{yx}(a, b)$

$$\parallel f_{xyy} = f_{yx y} = f_{yyx} \parallel$$

Ex: Let z be defined *implicitly* by $x^3 + y^3 + z^3 + 6xyz = 1$ (*).

Find $\frac{\partial z}{\partial x}$, $\frac{\partial z}{\partial y}$.

Sol: apply $\frac{\partial}{\partial x}$ to (*), keeping y constant:

$$3x^2 + 3z^2 \frac{\partial z}{\partial x} + 6yz + 6xy \frac{\partial z}{\partial x} = 0$$

$$\Rightarrow \frac{\partial z}{\partial x} = - \frac{3x^2 + 6yz}{3z^2 + 6xy} = - \frac{x^2 + 2yz}{z^2 + 2xy}$$

apply $\frac{\partial}{\partial y}$ to (*), keeping x constant:

$$3y^2 + 3z^2 \frac{\partial z}{\partial y} + 6xz + 6xy \frac{\partial z}{\partial y} = 0$$

$$\Rightarrow \frac{\partial z}{\partial y} = - \frac{y^2 + 2xz}{z^2 + 2xy}$$

Ex: $f(x, y, z) = \sin(3x + yz)$. Calculate $f_{xxyz} = \frac{\partial^4 f}{\partial x^2 \partial y \partial z}$

Sol: $f_x = 3 \cos(3x + yz)$

$$f_{xx} = -3 \sin(3x + yz)$$

$$f_{xxy} = -3z \cos(3x + yz)$$

$$f_{xxyz} = 3yz \sin(3x + yz) - 3 \cos(3x + yz)$$

Laplace equation: $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ (*)

2 partial differential equation

Ex: Check that $u(x, y) = e^x \sin y$ is a solution of (*).

Sol:

$$\left. \begin{array}{l} u_x = e^x \sin y \\ u_y = e^x \cos y \end{array} \right\} \Rightarrow u_{xx} + u_{yy} = 0$$
$$\left. \begin{array}{l} u_{xx} = e^x \sin y \\ u_{yy} = -e^x \sin y \end{array} \right\}$$

The Chain Rule:

Reminder: $y = f(x)$, $x = g(t)$ then y is indirectly a function of t

and $\frac{dy}{dt} = \frac{dy}{dx} \cdot \frac{dx}{dt} = f'(x) \cdot g'(t) = f'(g(t)) \cdot g'(t)$ [Case 0]

Case 1: $z = f(x, y)$, $x = g(t)$, $y = h(t)$

then
$$\frac{dz}{dt} = \frac{\partial f}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial f}{\partial y} \cdot \frac{dy}{dt}$$
$$\left(\text{or } \frac{\partial z}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial z}{\partial y} \cdot \frac{dy}{dt} \right)$$

Ex: $z = \sqrt{1+xy}$, $x = \sin t$, $y = t^2$. a) Find $\frac{dz}{dt}$ b) Find $\frac{dz}{dt}$ at $t = \pi$

Sol: (a)
$$\frac{dz}{dt} = \frac{\partial z}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial z}{\partial y} \cdot \frac{dy}{dt} = \frac{y}{2\sqrt{1+xy}} \cos t + \frac{x}{2\sqrt{1+xy}} 2t$$
$$= \frac{\frac{t^2}{2} \cos t + t \sin t}{\sqrt{1+t^2 \sin t}}$$

(b) $t = \pi \Rightarrow x = 0, y = \pi^2 \leadsto -\frac{\pi^2}{2}$

Case 2: $z = f(x, y)$ and $x = g(s, t)$, $y = h(s, t)$

then $\frac{\partial z}{\partial s} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial s} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial s}$ and $\frac{\partial z}{\partial t} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial t} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial t}$

Ex: $z = (x - y)^5$, $x = s^2 t$, $y = s t^2$. Find $\frac{\partial z}{\partial s}$, $\frac{\partial z}{\partial t}$.

Sol:

$$\begin{aligned} \frac{\partial z}{\partial s} &= \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial s} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial s} = 5(x - y)^4 \cdot 2st - 5(x - y)^4 t^2 \\ &= 5(s^2 t - s t^2)^4 (2st - t^2) \end{aligned}$$

$$\begin{aligned} \frac{\partial z}{\partial t} &= \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial t} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial t} = 5(x - y)^4 s^2 - 5(x - y)^4 2st \\ &= 5(s^2 t - s t^2)^4 (s^2 - 2st) \end{aligned}$$

General case of Chain Rule:

$$u = u(\underbrace{x_1, \dots, x_n}_{n \text{ variables}}) \text{ and for each } j: x_j = x_j(\underbrace{t_1, \dots, t_m}_{m \text{ variables}})$$

then

$$\frac{\partial u}{\partial t_i} = \underbrace{\frac{\partial u}{\partial x_1} \frac{\partial x_1}{\partial t_i} + \dots + \frac{\partial u}{\partial x_n} \frac{\partial x_n}{\partial t_i}}_{n \text{ terms}}$$

Ex: $u = x + yz$, $x = t$, $y = \ln t$, $z = \sin t$

$$\begin{aligned} \frac{du}{dt} &= \frac{\partial u}{\partial x} \frac{dx}{dt} + \frac{\partial u}{\partial y} \frac{dy}{dt} + \frac{\partial u}{\partial z} \frac{dz}{dt} = 1 \cdot 1 + z \frac{1}{t} + y \cos t \\ &= 1 + \frac{\sin t}{t} + \ln t \cos t \end{aligned}$$

Implicit differentiation: If $y = f(x)$ is defined implicitly,
by $F(x, y) = 0$

We can find $\frac{dy}{dx}$ via $0 = \frac{d}{dx} F(x, y) = \frac{\partial F}{\partial x} \underbrace{\frac{dx}{dx}}_{=1} + \frac{\partial F}{\partial y} \frac{dy}{dx}$

$$\Rightarrow \frac{dy}{dx} = - \frac{\partial F / \partial x}{\partial F / \partial y} = - \frac{F_x}{F_y}$$

Ex: y defined by $x^2 + y^3 = 1$ Find $y'(x)$.

Sol: $F(x,y) = x^2 + y^3 - 1 \Rightarrow \frac{dy}{dx} = -\frac{F_x}{F_y} = -\frac{2x}{3y^2}$

Ex: $z = z(x, y)$ defined by $x^2 + y^5 + z^3 = 0$ ^(*). Find $\frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}$

Sol: $\frac{\partial}{\partial x} (*) : 2x + 0 + 3z^2 \frac{\partial z}{\partial x} = 0 \Rightarrow \frac{\partial z}{\partial x} = -\frac{2x}{3z^2}$

$$\frac{\partial}{\partial y} (*) : 0 + 5y^4 + 3z^2 \frac{\partial z}{\partial y} = 0 \Rightarrow \frac{\partial z}{\partial y} = -\frac{5y^4}{3z^2}$$